

# Shu Quan

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Google Scholar

GitHub

## Education

**Peking University, Shenzhen Graduate School**

Shenzhen, China

*M.S. in Environmental Science, GPA: 3.61/4.0*

*Sep. 2024 – Jun. 2027 (Expected)*

- Research focus: trustworthy AI, especially LLM/agent evaluation and safety; advisor: Prof. Yaodong Yang.
- Visiting research: LLM hallucination mitigation at HKUST(GZ) NLP-LAB; advisor: Prof. Xuming Hu.

**Henan University**

Kaifeng, China

*B.S. in Geographic Information Science, GPA: 3.68/4.0, Rank: 1/46*

*Sep. 2020 – Jun. 2024*

- Relevant coursework: Programming, Data Structures, Database Systems, Probability and Statistics.

## Research Interests

Trustworthy AI; LLM/Agent Evaluation and Safety; Medical AI; AI for Science.

## Publications and Preprints

Equal contribution is denoted by \*.

1. **A Blind Spot in Alignment: Quantifying Biosecurity Risks in Large Language Models**  
Shu Quan<sup>\*</sup>, Tianfang Hao<sup>\*</sup>, Sitong Fang, He Geng, Jiayi Zhou, Boyuan Chen, Kaile Wang, Donghai Hong, Juntao Dai, Yaodong Yang, Jiaming Ji  
*Under review, 2026*
2. **SafeCLI: Exploring System-Centric Risks in CLI Agents via Augmented Observability**  
Donghai Hong<sup>\*</sup>, Changhao Ju<sup>\*</sup>, Shu Quan<sup>\*</sup>, Yuyang Ren<sup>\*</sup>, Boyuan Chen, Jiayi Zhou, Kaile Wang, Juntao Dai, Yaodong Yang, Jiaming Ji  
*Under review, 2026*
3. **Dense Clinical Contrasts Enhance Medical Knowledge Updating in Large Language Models**  
Yangmin Huang<sup>\*</sup>, Shu Quan<sup>\*</sup>, He Geng, Xin Ye, Qianyun Du, Zhiyang He, Jiaxue Hu, Xiaodong Tao  
*Under review, 2026*
4. **SLM-MATRIX: A Multi-Agent Trajectory Reasoning and Verification Framework for Enhancing Language Models in Materials Data Extraction**  
Xin Li<sup>\*</sup>, Zhixuan Huang<sup>\*</sup>, Shu Quan, Cheng Peng, Xiaoming Ma  
*npj Computational Materials, 2025*
5. **Nonlinear Effects of Blue-Green Space Variables on Urban Cold Islands in Zhengzhou Analyzed with Random Forest Regression**  
Shu Quan<sup>\*</sup>, Maojuan Li<sup>\*</sup>, Tianqi Li, Haodong Liu, Yaohui Cui, Miaohan Liu  
*Frontiers in Ecology and Evolution, 2023*
6. **Analysis of the PM<sub>2.5</sub>–O<sub>3</sub> Pollution Characteristics and Its Potential Sources in Major Cities in the Central Plains Urban Agglomeration from 2014 to 2020**  
Shu Quan, Miaohan Liu, Boxuan Chen, Yuehua Huang, Meijuan Wang, Qingxia Ma, Yan Han  
*Atmosphere, 2023*

Research Experience

**Peking University, PKU-Alignment Group**  
*Research Assistant; advisor: Prof. Yaodong Yang*

Sep. 2024 – Present  
Beijing, China

- Built evaluation infrastructure for hidden failures in LLM and agent systems, including functional biosecurity risk (SPIKE-Bench), OS-level CLI-agent harms (SafeCLI), and layer-wise interpretability probes for physical properties in video VAE latents.
- Developed benchmarks, OS-level monitoring signals, and probing diagnostics that turn failures missed by surface metrics into observable evidence.

**Alibaba Health**  
*Research Intern, Tool-Use Evaluation for Medical LLM Agents*

Apr. 2026 – Jun. 2026  
Beijing, China

- Designed trajectory-aware diagnostics for medical LLM agents, decomposing tool use into tool selection, argument validity, evidence grounding, and pre-response verification across guideline lookup, drug-interaction checking, and triage routing; flagged under-calling, wrong-tool, invalid-argument, and ungrounded-evidence failures missed by final-answer accuracy.

**OPPO AI Center, Large Model Algorithm Department**  
*Research Intern*

Jun. 2025 – Oct. 2025  
Shenzhen, China

- Built evaluation protocols for DeepResearch long-form generation, covering planning quality, numerical claims, citation accuracy, source matching, and code-execution checks; factual accuracy improved from 60% to 78.5%.

**iFLYTEK, South China Research Institute**  
*Research Intern*

Feb. 2025 – May 2025  
Guangzhou, China

- Contributed early data and evaluation construction for medical LLM knowledge updating, turning guideline updates and SEER oncology cases into temporally anchored evaluation/supervision data later used in the SEER-Bench workflow.

**Henan University**  
*Undergraduate Researcher, Geospatial and Environmental Modeling*

2022 – 2023  
Kaifeng, China

- Conducted remote-sensing and statistical analyses of urban cold islands and atmospheric pollution, leading to first/co-first authored environmental science publications.

Honors and Awards

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|-----------------------------------------------------------------------------------|-----------|
| • Kaggle Silver Medal, LLM Adversarial Attack Competition, Rank 76/1,692 (Top 5%) | Feb. 2025 |
| • Outstanding Graduate, Henan Province                                            | Jun. 2024 |
| • National Scholarship, China (Top 1%)                                            | Sep. 2023 |

Technical Skills

**Programming:** Python, PyTorch, Hugging Face Transformers, vLLM, LangGraph, Linux, Git  
**LLM methods:** SFT, LoRA, reward modeling, GRPO, benchmark design, LLM-as-judge, multi-agent systems